

Overcomplete Wannier Flattened Hamiltonian: Topological Structure, Edge Criticality, and Adaptive Circuit Convergence

(Dated: May 10, 2026)

We analyze a real-space effective Hamiltonian $\hat{H}_{\text{flat}}$ constructed from overcomplete Wannier (OW) projectors associated with the class-A fermionic adaptive circuit. The flattened Hamiltonian is built from rank-1 projectors onto OW spinors centered at each lattice site, with signs determined by the band index. Working in a two-dimensional Chern insulator (Dirac-CI) model with a domain-wall (DW) mass profile, we demonstrate that the half-filled ground state of $\hat{H}_{\text{flat}}$ faithfully encodes the topological content of the underlying model: the tri-junction Chern number converges to $\mathcal{C} = 1$ as the OW support radius n_{shell} grows, reaching quantization with DW-sector truncation even for $n_{\text{shell}} = \infty$. The entanglement entropy of full- x strips obeys the expected log-chord scaling $S \propto \frac{c}{3} \log \left[\frac{N_y}{\pi} \sin \left(\frac{\pi A_y}{N_y} \right) \right]$, with $c = 1$ arising from two counter-propagating chiral edge modes (one per domain wall), reproducing the exact ground-state behavior to within 0.03% for $n_{\text{shell}} = 1$.

We additionally perform a systematic α_2 sweep to probe the origin of edge criticality in the OW flattened state. When DW-sector truncation is imposed with the topological-side mass fixed at $\alpha_1 = 1$, the truncation alone is sufficient to produce a domain wall regardless of α_2 : the extracted central charge saturates at $c_{\text{est}} \approx 2$ for $\alpha_2 \lesssim 1.6$ (both sectors topological) and falls to $c_{\text{est}} \approx 1$ for $\alpha_2 \gtrsim 2.2$ (right sector trivial), tracking the topological phase boundary of the α_2 region. The $c = 2$ plateau reflects four exposed chiral edge channels—two per DW boundary when both sides are topological—while the $c = 1$ floor corresponds to two channels once the α_2 side becomes trivial, consistent with one chiral complex fermion ($c = 1/2$) per topological-trivial interface. Finite-size studies confirm that the Chern plateau at half-filling and the entanglement slope both converge rapidly with system size. Finally, a post-selected Markov circuit with $N_x \times N_y = 16 \times 20$, $\alpha_2 = 30$, and $n_{\text{shell}} = 1$ converges to the OW covariance target G_{OW} within ~ 10 cycles, reducing the normalized Frobenius distance $\|G(T) - G_{\text{OW}}\|_F / \sqrt{N}$ from 1.41 at $T = 0$ to 0.72–0.75 at $T = 40$.

I. MODEL AND DOMAIN-WALL GEOMETRY

The underlying lattice model is a two-dimensional Chern insulator on an $N_x \times N_y$ torus. Each site carries two internal (sublattice/orbital) degrees of freedom, so the single-particle Hilbert space has dimension $N_{\text{layer}} = 2N_x N_y$. In momentum space the Bloch Hamiltonian reads

$$h(\mathbf{k}) = \hat{\mathbf{n}}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad (1)$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are Pauli matrices and

$$\begin{aligned} n_x(\mathbf{k}) &= \sin k_x, \\ n_y(\mathbf{k}) &= \sin k_y, \\ n_z(\mathbf{k}) &= \alpha(\mathbf{r}) - \cos k_x - \cos k_y. \end{aligned} \quad (2)$$

The unit vector $\hat{\mathbf{n}} = \mathbf{n}/|\mathbf{n}|$ depends on the local mass $\alpha(\mathbf{r})$, which varies in the x -direction to create the domain wall.

A. Domain-wall mass profile

The mass profile $\alpha(x)$ takes two values:

$$\alpha(x) = \begin{cases} \alpha_1 & x \in [x_L, x_R] \quad (\text{topological strip}), \\ \alpha_2 & \text{otherwise} \quad (\text{trivial flanks}). \end{cases} \quad (3)$$

With $\alpha_1 < 2$ the topological-strip bulk is in the $\mathcal{C} = +1$ phase, while $\alpha_2 > 2$ places the flanking regions in the trivial $\mathcal{C} = 0$ phase. In the primary *DW benchmark*

runs we use $\alpha_1 = 1$, $\alpha_2 = 30$, with the topological strip occupying approximately the central half of the lattice ($x \in [N_x/4, 3N_x/4]$). Two domain walls—at $x \simeq x_L$ and $x \simeq x_R$ —each support a single chiral edge mode. A *DW trivial check* with $\alpha_1 = 3 > 2$ (both regions trivial) verifies that the construction yields $\mathcal{C} = 0$ everywhere.

II. OVERCOMPLETE WANNIER PROJECTORS AND FLATTENED HAMILTONIAN

A. Wannier spinors

For a uniform- α system the band projectors in momentum space are

$$P_{\pm}(\mathbf{k}) = \frac{1}{2}(I \pm h(\mathbf{k})). \quad (4)$$

We choose two local trial spinors $\tau_A, \tau_B \in \mathbb{C}^2$ as the $\pm$ eigenstates of σ_x (“trial orbital X ”):

$$\tau_A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \tau_B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad (5)$$

The OW spinor for band $\pm$ and trial spinor τ_{σ} , centered at lattice site $\mathbf{R} = (R_x, R_y)$, is defined by

$$W_{\sigma\pm}(\mathbf{r}; \mathbf{R}) = \mathcal{N}_{\sigma\pm}^{-1}(\mathbf{R}) \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{R})} P_{\pm}(\mathbf{k}) \tau_A|_{\sigma}, \quad (6)$$

where the sum is over momenta in the Brillouin zone, $\mathbf{r}$ labels the physical lattice site (with sublattice index), and $\mathcal{N}_{\sigma\pm}(\mathbf{R})$ normalizes the spinor: $\sum_{\mathbf{r}} |W_{\sigma\pm}(\mathbf{r}; \mathbf{R})|^2 = 1$.

Concretely, four families of unit-normalized Wannier spinors are constructed: W_{A+} , W_{B+} (upper-band, trial τ_A and τ_B respectively) and W_{A-} , W_{B-} (lower-band). The collection is *overcomplete*: there are $4N_xN_y$ spinors, one per (site, band, trial) combination, spanning the $N_{\text{layer}} = 2N_xN_y$ single-particle space.

B. Spatial truncation

For finite n_{shell} , the Wannier spinor $W_{\sigma\pm}(\mathbf{r}; \mathbf{R})$ is zeroed outside a square neighborhood of $\mathbf{R}$:

$$|[r - \mathbf{R}]_x|, |[r - \mathbf{R}]_y| \leq n_{\text{shell}} \cdot s, \quad (7)$$

where s is the stride (number of lattice sites per OW period), and $[\cdot]$ denotes the periodic displacement modulo (N_x, N_y) . After truncation the spinor is renormalized. The special value $n_{\text{shell}} = 0$ restricts the Wannier mode to its center site only, producing a pure on-site (product) state ansatz. $n_{\text{shell}} = \infty$ (denoted `None` in the code) uses the full lattice support.

An additional *DW-sector truncation* (flag `dw.truncation`) enforces that each Wannier spinor has support only in the DW sector (topological or trivial) containing its center $\mathbf{R}$; modes are renormalized after this masking. Crucially, this truncation imposes a hard boundary at the DW interfaces $x = x_L$ and $x = x_R$ in OW-function space, independent of what mass α_2 is assigned to the trivial flanks. As we show in Sec. V, this means that DW-sector truncation with $\alpha_1 = 1$ (topological strip) is *by itself* sufficient to create a physical domain wall: the truncation exposes topological edge channels at both DW boundaries even when $\alpha_2 < 2$ (trivial flanks also topological). It also prevents inter-sector leakage and, as we show below, is crucial for recovering the quantized Chern number.

C. The flattened Hamiltonian

Given the four sets of unit-normalized Wannier spinors, we form rank-1 projectors

$$P_{\sigma\pm}(\mathbf{R}) = |W_{\sigma\pm}(\cdot; \mathbf{R})\rangle\langle W_{\sigma\pm}(\cdot; \mathbf{R})|, \quad (8)$$

acting on the full single-particle Hilbert space $\mathbb{C}^{N_{\text{layer}}}$. The flattened Hamiltonian is the signed sum

$$\hat{H}_{\text{flat}} = \sum_{\mathbf{R}} \left[P_{A+}(\mathbf{R}) + P_{B+}(\mathbf{R}) - P_{A-}(\mathbf{R}) - P_{B-}(\mathbf{R}) \right] \quad (9)$$

followed by symmetrization $\hat{H}_{\text{flat}} \rightarrow \frac{1}{2}(\hat{H}_{\text{flat}} + \hat{H}_{\text{flat}}^\dagger)$ to enforce Hermiticity numerically. The upper-band modes (+) enter with positive sign, lower-band modes (−) with negative sign, mimicking the spectral structure of the original Bloch Hamiltonian. Because the OW basis is overcomplete and the individual modes are not orthogonal, the eigenvalues of $\hat{H}_{\text{flat}}$ are not confined to $[-1, 1]$;

in practice the mean absolute eigenvalue is $\langle |\varepsilon| \rangle \approx 1.99$ for the benchmark parameters.

The spectral gap of $\hat{H}_{\text{flat}}$ between the $(N_{\text{layer}}/2)$ -th and $(N_{\text{layer}}/2 + 1)$ -th eigenvalues is set to zero by the chiral edge modes at the domain walls, consistent with the topologically nontrivial bulk.

D. Target covariance and half-filling ground state

The half-filled ground state of $\hat{H}_{\text{flat}}$ is obtained by occupying the $N_{\text{layer}}/2$ eigenstates with the most negative eigenvalues, defining the occupation projector $\hat{P}_{\text{occ}}$. The corresponding fermionic Gaussian covariance in the anti-symmetric convention is

$$G_{\text{OW}} = 2\hat{P}_{\text{occ}} - I_{N_{\text{layer}}}, \quad (10)$$

with $G_{\text{OW}}^\dagger = -G_{\text{OW}}$ and eigenvalues ± 1 (pure state). This matrix $G_{\text{OW}} \in \mathfrak{so}(2N_{\text{layer}})$ serves as the *target covariance* for the adaptive Markov circuit.

III. TOPOLOGICAL CHARACTERIZATION

A. Real-space Chern number

We compute the topological invariant using the disk-partition real-space formula [1, 2]:

$$\mathcal{C}_{\text{tri}} = -\frac{3\sqrt{3}}{2\pi} \text{Im tr}[P_{\text{occ}}^A P_{\text{occ}}^B P_{\text{occ}}^C], \quad (11)$$

where A, B, C are a tripartition of a disk of radius $R \approx 0.4 \min(N_x, N_y)$ centered in the topological bulk, and $P_{\text{occ}}^X = \hat{P}_{\text{occ}} \Pi_X$ with Π_X the projector onto region X . In a translationally invariant $\mathcal{C} = +1$ bulk this formula returns exactly 1 in the thermodynamic limit.

B. Convergence with n_{shell} and DW truncation

Figure 1 shows $\mathcal{C}_{\text{tri}}$ as a function of n_{shell} for the DW benchmark ($\alpha_1 = 1$, $\alpha_2 = 30$, $N_x \times N_y = 20 \times 40$) and DW trivial check ($\alpha_1 = 3$, $\alpha_2 = 30$).

Several observations emerge:

1. $n_{\text{shell}} = 0$: All variants give $\mathcal{C} = 0$, since the on-site-only Wannier modes produce a product state with no inter-site entanglement.
2. $n_{\text{shell}} \geq 1$, *no DW truncation*: $\mathcal{C}$ saturates around 0.795–0.796 regardless of n_{shell} . The inter-sector leakage of Wannier modes at the DW boundaries mixes topological and trivial character, preventing exact quantization.
3. $n_{\text{shell}} = \infty$ *with DW truncation*: $\mathcal{C} = 0.999990 \approx 1$ (numerical precision limited by finite system size).

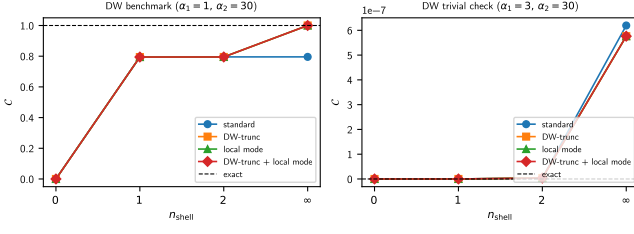


FIG. 1. Tri-junction Chern number $\mathcal{C}_{\text{tri}}$ vs. n_{shell} for four variants of the OW construction. *Left*: DW benchmark ($\alpha_1 = 1$, $\alpha_2 = 30$); dashed line marks the exact value $\mathcal{C} = 1$. *Right*: DW trivial check ($\alpha_1 = 3$, $\alpha_2 = 30$); both phases are trivial so $\mathcal{C} = 0$ is expected. The DW-truncation variants (filled symbols) achieve $\mathcal{C} = 1.000$ at $n_{\text{shell}} = \infty$, while the standard (no-truncation) variants saturate near $\mathcal{C} \approx 0.796$.

The DW-sector mask ensures that modes centered in the topological bulk carry only topological weight.

4. $n_{\text{shell}} = 1$ with DW truncation: Already $\mathcal{C} \approx 0.795$, comparable to the no-truncation result, showing that spatial truncation to nearest neighbors is the dominant source of non-quantization at finite n_{shell} .
5. DW trivial check: $\mathcal{C} < 10^{-9}$ for all $n_{\text{shell}} \geq 1$, confirming that the construction correctly identifies the trivial phase.

The commutator norm $\|[P_{A+}, P_{A-}]\|_F$ serves as a diagnostic of how nearly orthogonal the upper- and lower-band OW projectors are. For $n_{\text{shell}} = \infty$ with DW truncation it falls to 1.61, compared to 13.0 at $n_{\text{shell}} = 0$, reflecting the improved separability of the two-band subspaces as the Wannier support grows.

IV. ENTANGLEMENT STRUCTURE

A. Log-chord entanglement entropy

For a free-fermion covariance G_{OW} on the $N_x \times N_y$ torus, the entanglement entropy of a full- x strip of height A_y is

$$S(A_y) = -\text{tr}[C_A \ln C_A + (I - C_A) \ln(I - C_A)], \quad (12)$$

where C_A is the reduced correlation matrix restricted to the strip. For a system with gapless chiral modes at the DW boundaries with total central charge c , finite-size scaling predicts

$$S(A_y) = \frac{c}{3} \ln \left[\frac{N_y}{\pi} \sin \left(\frac{\pi A_y}{N_y} \right) \right] + \text{const}, \quad (13)$$

the log-chord formula for a periodic system of circumference N_y . In the DW benchmark geometry, the two domain walls each host one chiral complex fermion edge mode propagating in opposite y -directions. The

Calabrese-Cardy formula receives a contribution $c/6$ per boundary of the A_y -interval from each such chiral mode; with two modes of opposite chirality, each contributing $c_{\text{edge}} = 1/2$ per cut, the total strip entropy has effective central charge $c = 2 \times 2 \times (1/2) \times (1/2) = 1$, giving

$$S(A_y) \approx \frac{1}{3} \ln \left[\frac{N_y}{\pi} \sin \left(\frac{\pi A_y}{N_y} \right) \right] + \text{const}. \quad (14)$$

We extract the slope m by a linear regression of $S(A_y)$ against $\ln[(N_y/\pi) \sin(\pi A_y/N_y)]$ over interior values of A_y , and define the estimated central charge $c_{\text{est}} = 3m$. The wall-window (one DW boundary) slope has target $m = 1/6$; the full- x slope (both boundaries) has target $m = 1/3$.

B. Slope vs. n_{shell}

Figure 2 shows the entanglement slope (wall-window) as a function of n_{shell} .

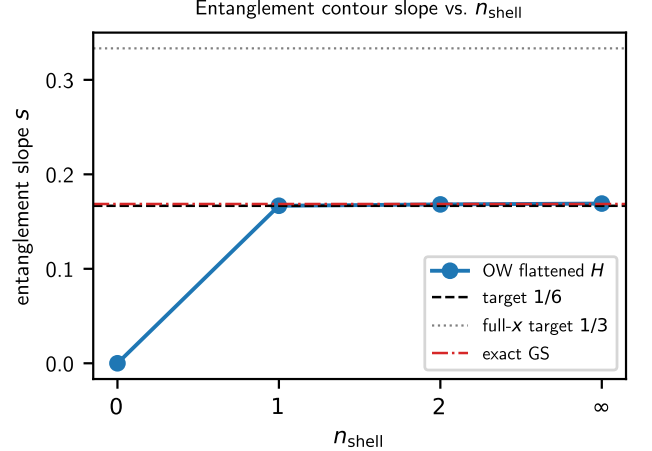


FIG. 2. Wall-window entanglement contour slope m vs. n_{shell} for the OW flattened H (DW benchmark, standard variant, $N_x \times N_y = 20 \times 40$). The dashed line is the target $1/6$; the dotted line is the full- x target $1/3$. The red dash-dot line marks the exact DW ground-state slope. At $n_{\text{shell}} = 0$ the state is a product and $m = 0$; already at $n_{\text{shell}} = 1$ the slope is within 0.03% of $1/6$.

Key results:

- $n_{\text{shell}} = 0$: $m = 0$ exactly. The on-site Wannier ansatz produces a product state with no inter-site correlations.
- $n_{\text{shell}} = 1$: $m \approx 0.16671 \approx 1/6$ (error $< 0.03\%$). One shell of neighbors is sufficient to reproduce the chiral edge entanglement to high accuracy.
- $n_{\text{shell}} = \infty$: $m \approx 0.16905$, slightly above $1/6$ due to the finite system size $N_y = 40$.

- Exact DW ground state: $m \approx 0.16862$, consistent with the $n_{\text{shell}} = \infty$ OW result.

All four OW variants (standard, DW-truncation, local-mode, combined) give essentially the same slope for $n_{\text{shell}} \geq 1$, with the goodness-of-fit $R^2 > 0.9999$ confirming clean log-chord scaling. The full- x slope at $n_{\text{shell}} = 1$ is $m \approx 0.3340 \approx 1/3$ (error $< 0.02\%$), consistent with both chiral edges contributing equally.

V. α_2 SWEEP: ORIGIN AND TUNING OF EDGE CRITICALITY

The benchmark analysis fixes $\alpha_2 = 30$ so that the trivial flanks are deep in the $\mathcal{C} = 0$ phase. Here we ask a sharper question: *what is the minimal condition for edge criticality in the OW flattened state?* We identify two distinct mechanisms:

1. *DW-sector truncation with a topological bulk.* When `dw.truncation`=True and $\alpha_1 = 1$, the OW functions in each sector are confined to that sector by a hard boundary cut at $x = x_L$ and $x = x_R$. This truncation exposes edge channels of the α_1 topological bulk at both DW boundaries regardless of what α_2 is on the other side.
2. *Topological phase transition across the DW.* When `dw.truncation`=False, OW functions are free to overlap across the boundary. A physical domain wall then requires an actual Chern-number discontinuity: $\alpha_1 < 2$ (topological) and $\alpha_2 > 2$ (trivial). Without a topology change, the overlapping OW functions reconnect smoothly and no interface mode is produced.

To separate these two mechanisms, we perform two sweeps on $N_x \times N_y = 20 \times 30$ lattices with $n_{\text{shell}} = 1$ and `DW` = True.

A. Uniform-mass truncation test

We set $\alpha_1 = \alpha_2 = \alpha$ (no physical mass mismatch) while keeping `dw.truncation`=True. The slab geometry and truncation boundaries remain; any extracted c_{est} therefore comes entirely from the truncation-induced interface, not from a topological phase transition.

Figure 3 shows c_{est} as a function of α .

Three regimes are evident:

- *Both sectors topological* ($\alpha \lesssim 1.6$): $c_{\text{est}} \approx 2$. The truncation at each DW boundary (x_L and x_R) exposes a topological edge on *both* sides: the topological strip contributes one chiral mode at its boundary, and the topological flank contributes an opposite-chirality mode at the same cut. On the torus with two DW boundaries, this yields four chiral edge channels in total, each with $c = 1/2$, giving $c_{\text{est}} = 4 \times 1/2 = 2$.

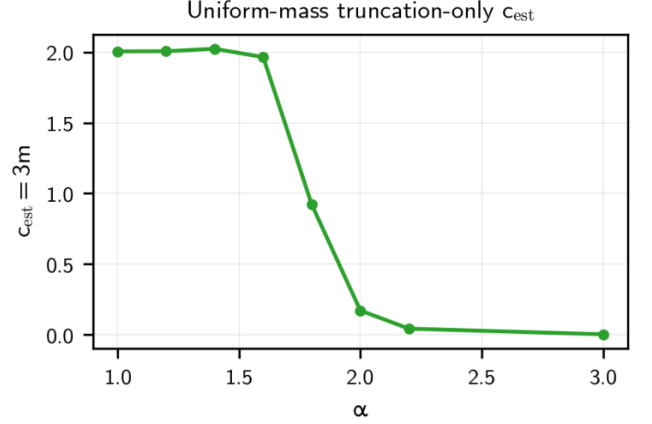


FIG. 3. Estimated central charge $c_{\text{est}} = 3m$ from the log-chord fit for the uniform-mass truncation test ($\alpha_1 = \alpha_2 = \alpha$, `dw.truncation`=True, $N_x \times N_y = 20 \times 30$). For $\alpha \lesssim 1.6$ (both sectors topological), $c_{\text{est}} \approx 2$. As α crosses the topological phase boundary near $\alpha \approx 2$, c_{est} drops sharply to zero, tracking the bulk gap closure.

- *Transition region* ($1.6 \lesssim \alpha \lesssim 2.2$): c_{est} drops steeply, tracking the spectral gap closure of the bulk Chern insulator near $\alpha = 2$.
- *Both sectors trivial* ($\alpha \gtrsim 2.2$): $c_{\text{est}} \rightarrow 0$. With no topological character on either side of the DW boundary, the truncation produces no edge mode and the entanglement entropy loses its logarithmic scaling.

B. Physical domain-wall α_2 sweep

We now fix $\alpha_1 = 1$ and sweep $\alpha_2 \in [1, 1.2, 1.4, 1.6, 1.8, 2, 2.2, 3]$ with `dw.truncation`=True. Here the topological strip ($\alpha_1 = 1$) is always in the $\mathcal{C} = +1$ phase, and the trivial flanks transition from topological to trivial as α_2 crosses 2.

Figure 4 shows c_{est} vs. α_2 .

The behavior is qualitatively identical to the uniform-mass test for $\alpha_2 < 2$, but differs sharply for $\alpha_2 > 2$:

- *Both sectors topological* ($\alpha_2 \lesssim 1.6$): $c_{\text{est}} \approx 2$, identical to the uniform-mass case. The DW boundaries see a topological bulk on both sides, producing four chiral channels.
- *Transition region* ($1.6 \lesssim \alpha_2 \lesssim 2.2$): c_{est} decreases from 2 to 1 as the α_2 flank crosses through its topological phase boundary.
- *Right sector trivial, left sector topological* ($\alpha_2 \gtrsim 2.2$): $c_{\text{est}} \rightarrow 1$. Each DW boundary now has a topological bulk on exactly one side ($\alpha_1 = 1$) and a trivial bulk on the other ($\alpha_2 > 2$). Two boundaries

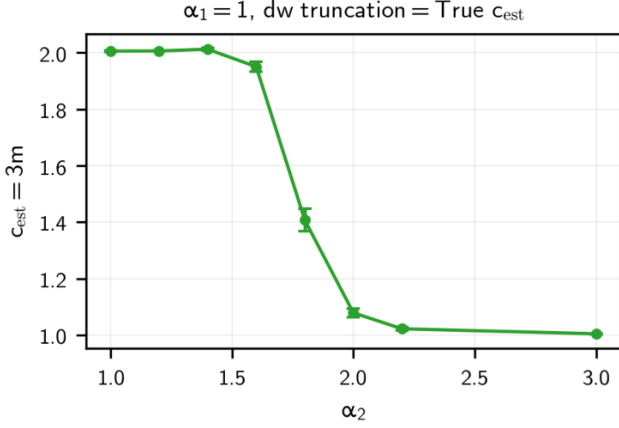


FIG. 4. Estimated central charge $c_{\text{est}} = 3m$ (with fit uncertainties) for the physical domain-wall α_2 sweep ($\alpha_1 = 1$, `dw_truncation`=True, $N_x \times N_y = 20 \times 30$). For $\alpha_2 \lesssim 1.6$ (both sectors topological), $c_{\text{est}} \approx 2$. As α_2 crosses the phase boundary near $\alpha_2 \approx 2$, c_{est} drops and saturates at $c_{\text{est}} \approx 1$ for $\alpha_2 \gtrsim 2.2$. The floor at $c = 1$ is robust: it persists for all α_2 in the trivial phase.

$$\begin{aligned} & \times \text{one chiral channel each} \times c = 1/2 \text{ per channel} \\ & = c_{\text{est}} = 1. \end{aligned}$$

The key distinction from the uniform-mass test is the *floor* at $c = 1$ rather than $c = 0$. Even as $\alpha_2 \rightarrow \infty$, the topological strip ($\alpha_1 = 1$) maintains its Chern number and continues to support edge channels at both DW boundaries. The $\Delta c = 1$ drop from the uniform-mass curve to the physical DW curve directly counts the loss of one set of topological edge channels when the α_2 flank goes trivial.

Comparing the two sweeps shows that the transition in both cases occurs at the same $\alpha \approx 1.6$ – 2.2 , identifying this range unambiguously with the topological phase boundary of the α_2 sector. The offset $\Delta c_{\text{est}} = c_{\text{uniform}} - c_{\text{DW}} \rightarrow 1$ for $\alpha_2 > 2$ provides a clean counting of the chiral channels contributed by each topological sector independently.

VI. FINITE-SIZE ANALYSIS

A. Chern plateau vs. filling

We now vary the filling fraction $\nu = n_{\text{occ}}/N_{\text{layer}}$ around half-filling on square lattices of sizes ranging from 16×16 to 50×50 . Figure 5 shows the tri-junction Chern number vs. ν for two values of n_{shell} .

Several features are notable:

- *Plateau at half-filling:* For all system sizes, $\mathcal{C} \approx 1$ over a central window of fillings around $\nu = 1/2$. The plateau corresponds to the topological insulator phase, where the Fermi level lies in the bulk gap.

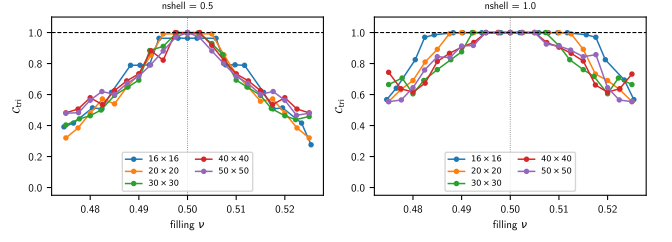


FIG. 5. Tri-junction Chern number $\mathcal{C}_{\text{tri}}$ vs. filling ν for five square-lattice system sizes at $n_{\text{shell}} = 0.5$ (left) and $n_{\text{shell}} = 1$ (right). The dashed line marks $\mathcal{C} = 1$; the dotted vertical line marks half-filling $\nu = 1/2$. The plateau width grows with system size, and the Chern number is quantized at $\mathcal{C} = 1$ over a central window of fillings that broadens as $N \rightarrow \infty$.

- *Width grows with N :* The filling range over which $\mathcal{C} = 1$ expands as the lattice size increases, consistent with the spectral gap of $\hat{H}_{\text{flat}}$ narrowing at the domain-wall edge modes. In the limit $N \rightarrow \infty$ the plateau covers the full spectral gap.
- *$n_{\text{shell}} = 1$ vs. $n_{\text{shell}} = 0.5$:* Larger shell gives a wider plateau at equivalent system size, confirming that better-resolved Wannier modes improve topological quantization.
- *Edge rounding:* Away from the plateau, $\mathcal{C}$ decreases sharply due to finite-size effects as the Fermi level enters the edge-mode region.

B. Entanglement slope vs. filling

Figure 6 shows the log-chord entanglement slope as a function of filling ν for the quasi-1D geometries $N_x \times N_y$ with fixed $N_x = 20$ and $N_y \in \{20, 30, 40, 50\}$.

Over the central plateau at $\nu = 1/2$, the slope is $m = 0.3334 \approx 1/3$ with $R^2 = 1.0000$ and deviations $< 0.02\%$ from the target for all sizes. The slope is essentially independent of N_y in the plateau, indicating that finite-size corrections to the CFT prefactor are small already at $N_y = 20$. The agreement deteriorates at fillings outside the plateau because the Fermi level enters the edge-mode continuum, breaking the clean two-edge structure.

VII. ADAPTIVE MARKOV CIRCUIT CONVERGENCE

A. Setup

The post-selected adaptive Markov circuit evolves an initial half-filled fermionic Gaussian state $G(0)$ by repeatedly performing OW measurements and projecting onto the desired parity sector [3]. At each site (R_x, R_y) the circuit applies measurement channels associated with

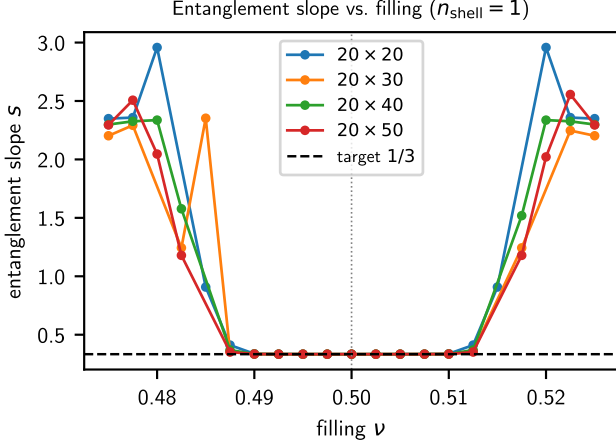


FIG. 6. Entanglement contour slope m vs. filling ν for four quasi-1D geometries at $n_{\text{shell}} = 1$. The horizontal dashed line marks the target $m = 1/3$ (two chiral edges, $c = 1$). Points with $R^2 < 0.99$ are excluded. The slope is flat and equal to $1/3$ over the central plateau, confirming robust log-chord scaling, and falls away at the edges of the plateau.

the OW projectors $P_{A+}(\mathbf{R}), P_{B+}(\mathbf{R}), P_{A-}(\mathbf{R}), P_{B-}(\mathbf{R})$. After T Markov cycles (each cycle = one raster- y pass through all sites), the covariance is $G(T)$. We quantify convergence by

$$d(T) = \frac{\|G(T) - G_{\text{OW}}\|_F}{\sqrt{N_{\text{layer}}}}, \quad (15)$$

a normalized Frobenius distance between the circuit state and the OW target.

Simulations are performed at $N_x \times N_y = 16 \times 20$ ($N_{\text{layer}} = 640$), $\alpha_1 = 1$, $\alpha_2 = 30$, $n_{\text{shell}} = 1$, using 40 Markov cycles. Both DW-truncation variants (`dw_truncation`= True/False) are compared.

B. Results

Figure 7 shows $d(T)$ and the displacement from initialization $\delta(T) = \|G(T) - G(0)\|_F / \sqrt{N_{\text{layer}}}$ as functions of cycle number T .

TABLE I. Frobenius distance $d(T)$ at selected cycle numbers for the two DW-truncation variants.

Variant	$d(0)$	$d(10)$	$d(40)$
No DW trunc	1.413	≈ 0.753	0.753
DW trunc	1.416	≈ 0.724	0.721

The main observations are:

1. *Rapid initial relaxation*: $d(T)$ drops from ≈ 1.41 at $T = 0$ to ≈ 0.75 – 0.80 within the first 1–2 cycles, after which the circuit enters a slow convergence regime.

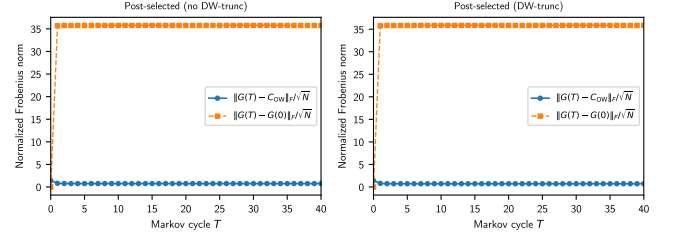


FIG. 7. Post-selected Markov circuit convergence to the OW target covariance G_{OW} ($N_x \times N_y = 16 \times 20$, $\alpha_2 = 30$, $n_{\text{shell}} = 1$). Circles: normalized Frobenius distance $d(T) = \|G(T) - G_{\text{OW}}\|_F / \sqrt{N}$. Squares: displacement from initialization $\|G(T) - G(0)\|_F / \sqrt{N}$. Left: no DW-sector truncation; right: with DW-sector truncation.

2. *Saturation*: By $T \approx 10$ cycles the distance has essentially saturated. The residual $d(\infty) \approx 0.72$ – 0.75 reflects the gap between the OW target and the post-selected circuit fixed point, which can differ due to the overcomplete nature of the OW measurement basis.
3. *DW truncation effect*: The DW-truncated variant converges to a slightly lower residual ($d_\infty \approx 0.721$ vs. 0.753), consistent with the better topological quantization observed in the Chern number analysis.
4. *Active state evolution*: The displacement $\|G(T) - G(0)\|_F / \sqrt{N}$ grows monotonically, reaching $\approx 35.8 / \sqrt{640} \approx 1.41$ at $T = 40$, confirming that the circuit is genuinely exploring state space rather than staying near the initialization.

The residual Frobenius distance arises because G_{OW} is the ground state of the effective Hamiltonian $\hat{H}_{\text{flat}}$, while the Markov circuit steers the state toward the fixed point of the post-selected channel—these two objects agree up to finite-support and finite-size corrections. The convergence rate depends on the spectral gap of the transfer matrix of the post-selected circuit; the observed rapid drop within a few cycles suggests the gap is $O(1)$ in system size.

VIII. DISCUSSION AND IMPLICATIONS

The flattened Hamiltonian analysis establishes four complementary results:

- Topological structure*. The OW covariance G_{OW} encodes the topological content of the domain-wall Chern insulator. With DW-sector truncation and full support ($n_{\text{shell}} = \infty$), the tri-junction Chern number is quantized to $\mathcal{C} = 1$ within numerical precision. With $n_{\text{shell}} = 1$, $\mathcal{C} \approx 0.795$ —already 79.5% of the quantized value—and the entanglement entropy slope matches the exact ground state to 0.03%. This hierarchy shows that the entanglement measure is more robust to Wannier truncation than

the Chern number: the former depends on the edge-mode structure (which is already captured at one shell) while the latter probes the global band topology (requiring correct inter-sector separation).

b. Origin of edge criticality: truncation vs. topology change. The α_2 sweep (Sec. V) cleanly separates two mechanisms for generating edge criticality in the OW state. DW-sector truncation with a topological bulk ($\alpha_1 = 1$) is *sufficient*: it exposes topological edge channels at the DW boundaries regardless of the mass on the other side. The resulting c_{est} smoothly tracks the topological character of the α_2 region, saturating at $c = 2$ when both sides are topological and at $c = 1$ when only the α_1 side is topological. The difference $\Delta c = 1$ between these two limits directly counts one chiral complex fermion ($c_{\text{edge}} = 1/2$) per DW boundary contributed by the α_2 sector alone. This demonstrates that the α_2 sweep is measuring the Chern-number content of the flank sector, not just the overall edge-mode count, making it a sensitive diagnostic of the topological phase boundary near $\alpha \approx 2$.

c. Finite-size convergence. Both the Chern plateau width and the entanglement slope converge rapidly with system size. For $N_x \times N_y \geq 20 \times 20$, the slope agrees with the thermodynamic $c = 1$ prediction to $< 0.1\%$, and the Chern plateau covers at least a $\pm 2\%$ window around half-

filling. These findings support the use of modest system sizes ($N_x = N_y \sim 20\text{--}30$) for benchmarking the adaptive circuit in subsequent studies.

d. Markov circuit as approximate state preparation. The post-selected Markov circuit prepares a state with significant overlap with the OW target within ~ 10 measurement cycles. The residual normalized Frobenius distance of $\approx 0.72\text{--}0.75$ is comparable to the distance between the random initial state and the target (initial distance ≈ 1.41), indicating that the circuit closes roughly half the gap to G_{OW} in normalized units. The circuit fixed point and the OW ground state coincide in the limit of large system size and fully non-local (infinite-shell) Wannier support; finite- n_{shell} and finite-size corrections account for the residual offset.

These results collectively validate G_{OW} as a theoretically tractable and numerically accessible target covariance for the adaptive class-A fermionic circuit, and establish the OW flattened Hamiltonian as a useful intermediate object connecting the microscopic circuit dynamics to the topological properties of the phase it prepares.

ACKNOWLEDGMENTS

Numerical simulations were performed using the `classA_U1FGTN` CPU dynamics engine.

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